Native long-XOR interface validation

Hypothesis: a fresh source-pinned wrapper can accept long XOR rows while retaining exact functional-primary models despite native internal cut variables. The unsupported implied_by entry point is disabled. No native core or Gaussian settings are changed.

The frozen worker passed. It enumerated all128 expected even-parity8-bit models, checked against all256 primary vectors by independent integer-popcount oracle. All256 declared AND outputs and the parity output were replayed. The265-variable XOR row was accepted; an empty true-parity row was rejected. Native statistics recorded265 declared and396 internal coordinates, with0 auxiliary decisions. This exercises internal coordinates on this fixture, not arbitrary long-row correctness.

Native integration test1/1 passed with its pinned library. It is ignored by default in ordinary CI. All26 native configuration readback values were exact. Wrapper/core hashes,159 core-source hashes, input contract and30-second/4GiB/CPU0 budget were frozen. The streaming compatibility coordinator verified the journal; the outer supervisor charged child startup and complete lifetime.

This is interface correctness evidence, not Gaussian admission, actual compressed-norm validation, a hard-root solve, novelty or a speed comparison. The single-row fixture initialized no Gaussian matrix. Original all-roots coverage and the matching genuine F4/F5/F6 end-to-end gate remain open. No release claim is made.

Source review found that add_every_combination_xor retains the original XOR but also introduces CNF cut coordinates. The265-variable fixture created131 such coordinates. A dense full-cone interface may therefore increase native work despite fewer external variables. Exact actual counts after cleaning remain unmeasured.

Next: first profile a cut-aware hybrid cone representation; then encode the canonical affine definitions and AND clauses of the exact norm compiler; compare complete tiny primary model sets, including zero/repeated sources, against the independent literal norm oracle. Only then freeze a new hard search. Do not retry the unchanged old timeout or infer speed from representation counts.

Publication review and explicit authorization extended this interaction beyond the nominal15-minute cycle. No scientific worker budget was enlarged. One new bounded interface experiment ran. All earlier attempts are preserved.

Evidence: research/native_long_xor_interface_20261017_v1/ and lab/ledger/attempts/native-long-XOR-interface-complete-controls-20261017-v1/0001/. Canonical performance graphs remain unchanged because no new qualified performance result exists.

8 primary bits; 265-variable parity row

256 declared AND outputs, all replayed

396 internal coordinates; 0 auxiliary decisions

128 complete models match independent exhaustive oracle

Interface validation only: no speed measurement